

AWS Backend & Frontend Deployment – Step-by-Step Guide

Overview

This document explains the **step-by-step process** to deploy a **Backend (FastAPI)** and **Frontend (React/Node build)** application on AWS using **EC2** and **S3**. It is written in a clear, practical format suitable for **project documentation, SOPs, or beginner-to-intermediate learners**.

Prerequisites

- AWS account with access to **EC2** and **S3**
 - EC2 instances already created:
 - `test_backend`
 - `test_frontend`
 - Backend and frontend ZIP files already uploaded to **S3**
 - EC2 instances configured with:
 - Internet access
 - AWS CLI installed and configured
 - Required security group rules (SSH, HTTP/HTTPS)
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BACKEND DEPLOYMENT (FastAPI)

Step 1: Login to AWS Console

1. Log in to the **AWS Management Console**
 2. Search for **EC2** from the services menu
 3. Click on **Instances**
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Step 2: Connect to Backend EC2 Instance

1. Select the instance `test_backend`
 2. Click **Connect**
 3. Connect using **EC2 Instance Connect / SSH**
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Step 3: Copy Backend ZIP from S3

1. Search for **S3** in AWS Console
2. Open the bucket containing the backend ZIP file
3. Select **backend.zip**
4. Copy the **S3 Object URI**

Step 4: Clean Old Backend Files (If Any)

Before copying the new backend code, remove any existing backend folders to avoid conflicts.

```
sudo rm -rf backend
```

Verify cleanup:

```
ls
```

Step 5: Download Backend ZIP in EC2

```
aws s3 cp <PASTE_S3_URI_HERE> .
```

Verify download:

```
ls
```

Step 5: Extract Backend Code

```
sudo unzip backend.zip
```

Navigate to project directory:

```
cd backend/fastapi/
```

Step 6: Configure Environment Variables

Create `.env` file:

```
cat <<EOF > .env
# Add environment variables here
EOF
```

Move back to backend root:

```
cd ..
```

Step 7: Create and Activate Virtual Environment

```
python3 -m venv venv  
source venv/bin/activate
```

Step 8: Install Backend Dependencies

```
cd fastapi/  
pip install -r requirements.txt
```

Step 9: Restart Backend Service using PM2

Check PM2 status:

```
pm2 status
```

Restart backend process:

```
pm2 restart 0
```

FRONTEND DEPLOYMENT

Step 10: Connect to Frontend EC2 Instance

1. Go to **EC2 → Instances**
 2. Select `test_frontend`
 3. Click **Connect** and access the instance
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Step 11: Copy Frontend ZIP from S3

1. Open **S3** service
2. Select the frontend ZIP file
3. Copy the **S3 Object URI**

Step 12: Navigate to Web Root Directory

```
cd /var/www/html/  
ls
```

Step 13: Clean Old Frontend Files (If Any)

Before copying the new frontend build, remove the existing frontend folder.

```
sudo rm -rf frontend
```

Verify cleanup:

```
ls
```

Step 14: Download Frontend ZIP

```
cd /var/www/html/  
ls
```

Step 13: Download Frontend ZIP

```
sudo aws s3 cp <PASTE_S3_URI_HERE> .
```

Verify file:

```
ls
```

Step 14: Extract Frontend Code

```
unzip frontend.zip  
cd frontend
```

Step 15: Install Dependencies and Build Frontend

```
npm install  
npm run build
```

Step 16: Restart Nginx Server

```
sudo systemctl restart nginx
```

Verification Checklist

- Backend API accessible via public IP / domain
 - Frontend UI loads correctly in browser
 - PM2 shows backend process as **online**
 - Nginx service running successfully
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Notes & Best Practices

- Always verify AWS IAM permissions for S3 access
 - Use environment variables instead of hardcoding secrets
 - Ensure security groups allow required ports (80, 443, backend port)
 - Keep separate instances for frontend and backend for scalability
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Document Type: Deployment SOP

Environment: AWS EC2 + S3

Backend: FastAPI (PM2)

Frontend: Node / React (Nginx)